THE COOPER UNION FOR THE ADVANCEMENT OF SCIENCE AND ART
ALBERT NERKEN SCHOOL OF ENGINEERING

Computer Vision for Vehicle Emission Estimation

By

Aidan Cusa

A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Engineering

Advisor

Carl Sable

THE COOPER UNION FOR THE ADVANCEMENT OF SCIENCE AND ART

ALBERT NERKEN SCHOOL OF ENGINEERING

This thesis was prepared under the direction of Carl Sable and has received approval. It was submitted to the Dean of the School of Engineering and the full Faculty, and was approved as partial fulfillment of the requirements for the degree of Master of Engineering.

Barry L. Shoop, Ph.D, P.E. - Date
Dean, Albert Nerken School of Engineering

Carl Sable, Ph.D - Date
Candidate's Thesis Advisor

Acknowledgements

I would like to express my sincere gratitude to my mentor Professor Carl Sable for his guidance, patience, and encouragement throughout this project. I would also like to thank my teammates, Sun Kongsonthana, and Minahil Bakhtawar for their collaboration, creativity, and teamwork during the development of the Remote Micro-Robot middleware.

I am grateful to the College of Design and Engineering at the National University of Singapore for providing the resources and support that made this work possible. Finally, I would like to thank The Cooper Union for providing me with this incredible opportunity to represent the school abroad.

Abstract

Hello

Contents

1 Introduction **1**

1.1 Motivation 1

2 Related Work **3**

2.1 Traditional Air Quality Monitoring Methods 3

2.1.1 Stationary Air Quality Monitoring Stations (AQMS) 3

2.1.2 Mobile Monitoring Platforms 4

2.1.3 Low-Cost Sensors and On-Board Diagnostics (OBD) 4

2.2 Satellite-Based Remote Sensing 5

2.2.1 Resolution and Instrumentation 5

2.2.2 Physical and Environmental Constraints 5

2.2.3 The Dependency on Bottom-Up Inventories 5

3 Methodology **6**

3.1 Performance Metrics 6

3.1.1 Intersection over Union 6

3.1.2 Confusion Matrix 7

3.1.3 Precision-Recall Curve 8

3.1.4 Average Precision and mAP 10

References **12**

1 Introduction

1.1 Motivation

Urban air pollution remains one of the most critical public health and environmental challenges of the twenty-first century. The World Health Organization (WHO) estimates that approximately 99% of the global population breathes air containing pollutant concentrations that exceed guideline limits [1], [2]. In 2023 alone, air pollution contributed to 7.9 million deaths worldwide, with 86% of these attributed to noncommunicable diseases such as cardiopulmonary conditions and lung cancer [2]. Beyond its immediate health impacts, atmospheric pollution catalyzes broader environmental degradation. Short-lived climate pollutants and greenhouse gases contribute to global warming, weather variability, and the intensification of the water cycle, which in turn threatens infrastructure and coastal settlements [3], [4].

Road vehicles are the central driver of both the climate and health aspects of this crisis. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report notes that direct greenhouse gas (GHG) emissions from the transport sector accounted for 23% of global energy-related CO_2 emissions in 2019 [5]. Of these, 70% originated specifically from road vehicles [5]. Without intervention, transport emissions are projected to double by 2050, driven by increasing vehicle ownership in developing economies [6]. In urban environments, these emissions are particularly consequential because they are released at ground level along the very corridors where people live, work, and commute, creating immediate exposure risks [3].

New York City (NYC) serves as a critical intersection of dense population, heavy traffic activity, and measurable health burdens. As of July 2024, there are 8.48 million residents in all five boroughs, with (x amount of people near major roadway/highway that is high risk exposure. ASK SABLE: should I try to find the number for this if I have the time? Like should I try to contribute as much as I can? I can't find a number past 2007) [7]. While the typical passenger vehicle emits approximately 411 grams of CO_2 per mile [8], it also releases pollutants toxic to human health, most notably fine particulate matter ($PM_{2.5}$). This pollutant is described by the NYC Department of Health and Mental Hygiene as "the most harmful air pollution for humans to breathe" as $PM_{2.5}$ penetrates deep into the lungs and enters the bloodstream, circulates throughout the body, and causes systemic inflammation [9].

The local impact of these emissions is quantifiable and severe. In New York City, $PM_{2.5}$ levels from all sources contribute to approximately 2,000 deaths and 5,150 emergency department visits annually [9]. 14% of local $PM_{2.5}$ is directly attributable to everyday car, bus, and truck traffic, resulting in an estimated 320 premature deaths and 870 hospitalizations each year [9]. In New York State, the combined health and climate costs attributable to passenger vehicles totaled approximately $7.9 billion in 2015, with health-related costs comprising two-thirds of that total [10].

Despite the well-documented impacts of transportation emissions, effective mitigation is hindered by the difficulty of accurate, timely measurement. Emissions from road vehicles are not distributed uniformly across space or time, but rather concentrated in specific "hotspots," vary by time of day, and depend heavily on vehicle composition. For instance, heavy-duty vehicles contribute disproportionately to nitrogen oxides (NOx) and particulate matter relative to light-duty cars [11], while congestion increases per-mile emissions through idling and stop-and-go driving. Consequently, understanding not only total vehicle counts but also the specific composition of traffic flows is critical for estimating localized emissions burdens.

Traditional emissions estimators, however, often rely on aggregate fleet statistics and modeled travel demand applied at coarse spatial scales. While indispensable for long-term planning, these methods often lack the granularity required to evaluate street-level exposure disparities. Similarly, fixed air-quality monitoring stations measure ambient concentrations but cannot directly attribute pollutants to specific vehicle classes or immediate traffic patterns. Furthermore, monitoring infrastructure is unevenly distributed globally. High-income regions typically possess denser monitoring networks, whereas many low- and middle-income countries lack adequate coverage, leaving large populations without reliable local air quality data [12].

- can put better citations for here too

To address these limitations, this thesis proposes leveraging existing video infrastructure as a scalable monitoring tool. New York City maintains an extensive network of publicly accessible traffic cameras, offering a unique opportunity to generate high-resolution data without deploying new hardware. By applying computer vision techniques to identify vehicle classes and counts, combined with established emission factors, it is possible to generate spatially resolved estimates of traffic-related emissions. This approach does not replace traditional air-quality monitoring but complements it, providing a granular view of how traffic composition drives environmental health impacts. Moreover, this methodology

offers a potential blueprint for regions with limited monitoring resources, suggesting that computer vision-based estimation could become an accessible, scalable tool for global environmental health monitoring.

2 Related Work

2.1 Traditional Air Quality Monitoring Methods

This section reviews the evolution of methodologies for quantifying urban air pollution, ranging from traditional stationary infrastructure to deep learning approaches. It evaluates the strengths and limitations of direct measurement approaches. It identifies the critical data gaps in spatial resolution and source attribution that necessitate the development of high-resolution, vision-based bottom-up inventories.

Rewrite: Current approaches to urban air quality assessment are generally categorized into top-down and bottom-up methodologies. Top-down approaches infer emission fluxes from observed atmospheric concentrations, using data from satellite remote sensing or stationary monitoring networks to validate regional models. Conversely, bottom-up approaches calculate emissions at the source by combining activity data (e.g., traffic flow) with emission factors. This section reviews the evolution of these monitoring paradigms, evaluating the limitations of current top-down observation in resolving street-level heterogeneity and identifying the critical need for high-resolution, vision-based activity data to improve bottom-up inventories.

2.1.1 Stationary Air Quality Monitoring Stations (AQMS)

Regulatory-grade stationary networks are the standard for air quality assessment, providing high-quality, standardized data ideal for analyzing spatiotemporal trends across large regions [1]. However, the number of these fixed-site monitors are limited. For example, in U.S. urban areas with continuous monitoring, there are typically only 2-5 monitors per million people [2]. Consequently, they fail to capture hyperlocal, street-by-street variations in pollution, which is the scale at which human exposure and exceedances of ambient standards often occur [1], [2]. Research indicates that concentrations of pollutants such as ultrafine particles (UFP), black carbon (BC), and nitrogen oxides (NO_x) can vary by 5–8 times within individual city blocks and fluctuate

significantly over distances of less than 300 meters [1], [2]. While dispersion models and satellite remote sensing are often used to supplement stationary data, they face intrinsic resolution limitations (discussed further in Section 2.2). Thus, they cannot fully resolve the fine-scale gradients (10–300 m) driven by local traffic emissions [2].

2.1.2 Mobile Monitoring Platforms

To address the spatial limitations of fixed networks, mobile monitoring (mounting reference-grade instruments on fleet vehicles) has emerged as a method to capture high-resolution spatial data [1]. Notable implementations include equipping Google Street View cars to repeatedly sample urban roadways, successfully revealing persistent pollution patterns and stable "hotspots" attributable to local sources [2]. This approach allows for the identification of source impacts without a priori assumptions about emission rates [1].

However, while mobile monitoring maximizes spatial coverage, it lacks temporal continuity [1]. Because a vehicle cannot measure all locations simultaneously, data must be aggregated over long periods or modeled to estimate averages [1]. Furthermore, measurement locations are inherently biased toward the roadway (on-road), which can differ from off-road concentrations (e.g., at home addresses) by 20–30%, necessitating careful adjustment when estimating population exposure [1].

2.1.3 Low-Cost Sensors and On-Board Diagnostics (OBD)

A shifting paradigm in monitoring involves the deployment of portable, lower-cost sensors capable of reporting near real-time data [3]. These devices are generally categorized into gas-phase sensors (electrochemical or metal oxide) and particulate matter sensors (optical scattering) [3]. While these allow for denser deployment and community-based science, they suffer from significant data quality issues. As of April 2026, there are currently no commercially available direct-reading sensors for PM mass (relying instead on light-scattering surrogates) or specific hazardous air pollutants, and performance criteria for non-regulatory use remain undefined [3].

Similarly, the mandatory equipping of heavy-duty vehicles with On-Board Diagnostic (OBD) systems offers a potential stream of "big data" for real-time emission monitoring [4]. While this provides an opportunity to supervise in-use vehicles remotely, the data quality is often poor and lacks a comprehensive assessment, requiring substantial

computational infrastructure to process effectively [4].

2.2 Satellite-Based Remote Sensing

While sensor networks provide point-based measurements, satellite remote sensing offers a top-down approach to air quality monitoring, playing a critical role in developing emission policies and forecasting regional air quality [5], [6].

2.2.1 Resolution and Instrumentation

The landscape of satellite monitoring has shifted with the deployment of instruments like the Tropospheric Monitoring Instrument (TROPOMI) on the Sentinel-5P mission. TROPOMI provides daily global measurements at an unprecedented spatial resolution of approximately 3.5 $\times$ 5.5 km^2, allowing for the detection of trace gases (NO_2, O_3, CO) and their linkages to human activity [7], [8]. Complementary infrared sounders, such as IASI and AIRS, provide vertical profile data, though at coarser footprint resolutions (12 km) [9]. Despite these advancements, even "high-resolution" satellite data remains too coarse (¿3 km) to resolve the fine-scale gradients (10–300 m) driven by urban traffic emissions, effectively smoothing out the street-level heterogeneity that defines urban exposure [2], [10].

2.2.2 Physical and Environmental Constraints

Beyond spatial resolution, satellite retrieval is limited by physical constraints, particularly regarding diurnal variation. "Top-down" observation is highly sensitive to the planetary boundary layer height; in the morning, a shallow boundary layer can mask surface-level pollutants, reducing sensitivityexactly when traffic emissions are often highest [10]. Furthermore, retrieval algorithms struggle with environmental variables such as surface reflectivity and viewing geometry (zenith angles), which can introduce artifacts into the data, particularly at high latitudes or in complex urban terrains [10].

2.2.3 The Dependency on Bottom-Up Inventories

To bridge the gap between regional satellite data and street-level reality, researchers rely on chemical transport models (CTMs) and data assimilation techniques. However, the efficacy of these models is fundamentally limited by the quality of their inputs. Recent reviews

indicate that high-resolution CTMs require detailed emission inventories (better than 1 km resolution) to resolve small-scale dynamical features like urban heat islands and street canyon effects [10]. The current lack of such granular, dynamic ”bottom-up” inventories remains a bottleneck for effective satellite data assimilation, validating the need for improved terrestrial emission modeling [10], [11].

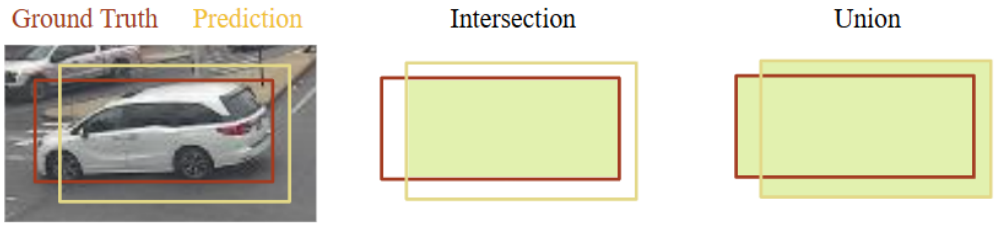
3 Methodology

3.1 Performance Metrics

To determine if using computer vision is an adequate method to estimate emissionscoefficients in New York City, the vehicle detection pipeline must be rigorously evaluated. Unlike standard image classification, object detection requires assessing two distinct tasks simultaneously: accurately localizing a vehicle within a complex urban environment and correctly classifying its emissions profile category (e.g., distinguishing a heavy-duty vehicle from a light-duty vehicle). This subsection details the metrics selected to quantify the model's performance. It begins by establishing the spatial evaluation criteria, builds toward aggregate threshold-agnostic metrics like mean Average Precision (mAP), and concludes by justifying these choices against traditional classification metrics that fall short in dynamic spatial contexts.

3.1.1 Intersection over Union

Before assessing the classification accuracy of the model, it is necessary to quantify the spatial accuracy of the predicted bounding boxes. This is achieved using the Intersection over Union (IoU) metric. IoU measures the overlap between a predicted bounding box and a ground truth bounding box. It is calculated as the area of their intersection divided by the area of their union.



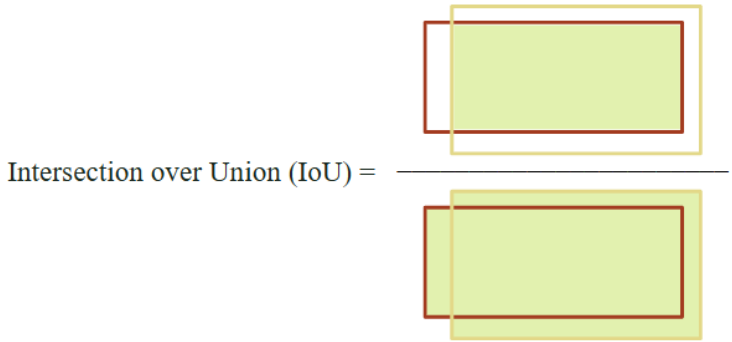


Figure 1: IoU Calculation

An IoU score ranges from 0 (no overlap) to 1 (perfect alignment). By establishing an IoU threshold (ex., 0.50), we define the strictness required for a spatial prediction to be considered valid.

3.1.2 Confusion Matrix

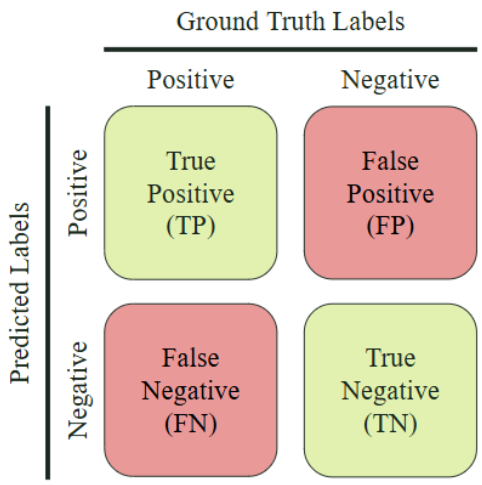


Figure 2: Confusion Matrix

With a defined IoU threshold, the traditional elements of a confusion matrix (Figure 2) can be adapted for object detection. Unlike pure image classification, where every image has a discrete label, object detection occurs in a continuous spatial domain. This is because bounding box coordinates can take on virtually infinite combinations across an image, so a model's spatial prediction will rarely align pixel-for-pixel with a human annotation. Thus, setting a threshold on the IoU metric acts as a spatial tolerance mechanism, converting

continuous coordinate overlaps into the discrete *True Positive* or *False Positive* categories required to calculate performance metrics. Within the context of this project, the entries of the confusion matrix correspond to:

- **True Positive (TP)**: A detection that matches a human-annotated ground truth box of the correct vehicle class with an IoU $\geq$ the defined threshold. For example, the model successfully draws a bounding box around a sedan and correctly classifies it as a Light-Duty Vehicle (LDV) with an IoU that exceeds the threshold.
- **False Positive (FP)**: A predicted box that fails either the spatial or classification criteria. In the context of urban traffic monitoring, this typically occurs in three ways:
  1. *Misclassification*: The model perfectly localizes an LDV but incorrectly classifies it as a Heavy-Duty-Vehicle (HDV), skewing the emissions profile for that frame.
  2. *Poor Localization*: The model correctly identifies a Bus, but the bounding box only captures a fraction of the vehicle, resulting in an IoU below the required threshold.
  3. *Background/Duplicate*: The model hallucinates a vehicle where there is only an empty patch of road, or it draws two separate bounding boxes over the same motorcycle.
- **False Negative (FN)**: A ground truth vehicle that the model failed to detect entirely, or localized so poorly that it was discarded. For instance, an LDV partially occluded by a building is completely ignored by the model, resulting in an uncounted emissions source.
- **True Negative (TN)**: Conceptually, a true negative represents every possible empty region in an image (such as patches of bare asphalt, sidewalks, buildings, or sky) where the model correctly predicted that *no* vehicle was present. Because the number of empty background regions in a dense urban scene is effectively infinite, True Negatives mathematically distort traditional evaluation metrics and are therefore excluded from object detection performance calculations.

3.1.3 Precision-Recall Curve

Using the *TP*, *FP*, and *FN* counts defined above, we can calculate Precision and Recall to evaluate the pipeline's ability and reliability in tracking urban traffic.

- **Precision** measures the reliability of the model's vehicle predictions.

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

  In the context of this project, out of all the bounding boxes the model predicted for a certain class (ex. Claiming to detect 50 HDVs), what fraction were correct? A high precision score ensures the model is not predicting false "ghost" vehicles or misclassifying vehicle types.

- **Recall** measures the model's ability to find vehicles in the frame.

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

  Out of all the actual ground truth objects present, what fraction did the model successfully detect? A high recall score ensures emission sources are not undercounted in the final analysis.

Modern object detectors like the DETR output a confidence score with each predicted bounding box, creating a trade-off between Precision and Recall. Lowering the confidence threshold allows the model to predict more vehicles (increasing Recall), but it simultaneously increases the risk of capturing background noise as valid vehicles (False Positives), thereby lowering Precision. By calculating Precision and Recall at every possible confidence threshold and plotting the results, we generate the Precision-Recall (PR) Curve.

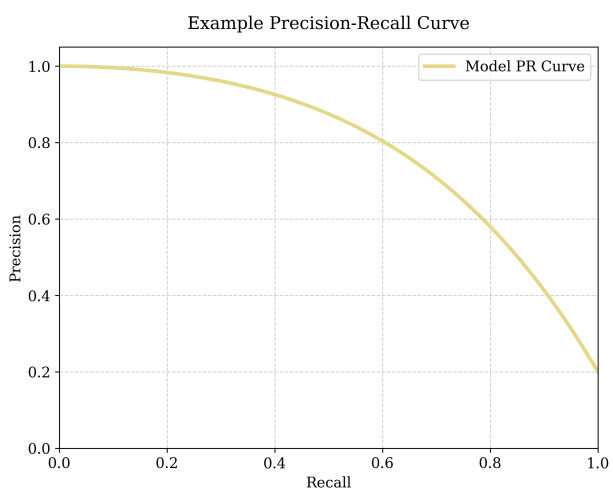


Figure 3: Example Precision-Recall Curve

The PR Curve illustrates the model's overarching stability and allows us to visualize the exact penalty in precision when we tune the model to capture a higher volume of traffic.

3.1.4 Average Precision and mAP

To quantify the trade-off depicted in the PR Curve into a single scalar value, we use the Average Precision (AP). AP represents the total area under the PR Curve, depicted below in Figure 4.

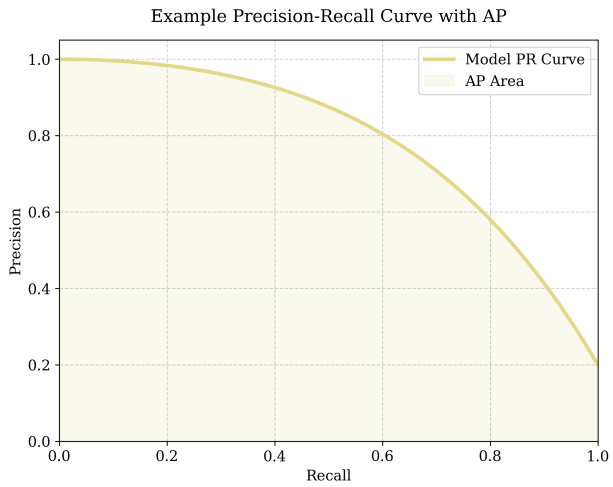


Figure 4: Average Precision within a Precision-Recall Curve

It represents the model's reliability across every possible confidence threshold for a single specific vehicle class (e.g., evaluating solely the model's ability to isolate Heavy-Duty Vehicles from the rest of the traffic flow). AP is calculated as follows:

$$AP = \int_0^1 p(r)dr \tag{3}$$

Where $p(r)$ is the precision at recall r. By calculating the area under the curve, AP penalizes models that suffer severe drops in precision when attempting to retrieve the hardest-to-find vehicles (such as heavily occluded trucks). To evaluate the air quality modeling pipeline across the entire vehicle, the individual AP scores for all distinct classes (LDV, MDV, HDV, Bus, and Two-Wheeler) are averaged to produce the Mean Average Precision (mAP):

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{4}$$

Where N is the total number of classes ($N = 5$ in this study). Because the mAP score relies on the initial spatial overlap requirement, standard evaluation protocols report performance at different Intersection over Union (IoU) thresholds. A common baseline is AP50 (AP calculated at an IoU threshold of 0.50), which tolerates moderate bounding box looseness. However, modern object detection systems evaluate performance using the protocol established by the Microsoft Common Objects in Context (MS COCO) dataset [13]. The COCO-style mAP (often denoted as $AP@[.50 : .05 : .95]$) averages the mAP across ten distinct IoU thresholds ranging from 0.50 to 0.95 with a step size of 0.05:

$$mAP_{COCO} = \frac{1}{10} \sum_{t \in T} mAP_t \tag{5}$$

Where $T = \{0.50, 0.55, 0.60, \ldots, 0.95\}$ and mAP_t is the Mean Average Precision evaluated at IoU threshold t. The primary objective of this thesis is accurately labeling vehicles for emissions modeling which makes the looser spatial tolerance of AP50 an intuitively appealing metric. However, this study utilizes the stricter COCO-style mAP as the primary performance indicator for several reasons:

- The proposed pipeline relies on cropping the detected bounding boxes to feed into a secondary classification model (see Section —). A loose bounding box that barely satisfies a 0.50 IoU may contain excessive background scene or partial features of nearby vehicles. This noise degrades the quality of the image crop, leading to downstream misclassification of the vehicle type (e.g., confusing an MDV with an HDV). Therefore, accurate vehicle classification depends on high quality localization.
- In heavily congested New York City traffic, relying solely on AP50 can hide localization failures. Sloppy bounding boxes that overlap multiple tailgating vehicles can still score under a 0.50 threshold, but will cause the system to merge distinct emission sources, leading to undercounting.
- Finally, COCO mAP is the standard in modern computer vision literature. By integrating across ten spatial strictnesses, it reduces the metric's sensitivity to any single arbitrary cutoff and prevents the model from artificially inflating its performance by outputting overly large regions of interest.

Utilizing the COCO mAP ensures that the detection architecture is not hallucinating large active regions, but is learning the boxes required to reliably isolate vehicles in urban environments.

References

[1] W. H. Organization. "Air pollution." (), [Online]. Available: https://www.who.int/health-topics/air-pollution (visited on 02/13/2026).

[2] H. E. Institute, "State of global air 2025: A report on air pollution and its role in the world's leading causes of death," Health Effects Institute, Boston, MA, 2025. [Online]. Available: www.stateofglobalair.org.

[3] E. Uherek, T. Halenka, J. Borken-Kleefeld, *et al.*, "Transport impacts on atmosphere and climate: Land transport," *Atmospheric Environment*, Transport Impacts on Atmosphere and Climate: The ATTICA Assessment Report, vol. 44, no. 37, pp. 4772–4816, Dec. 1, 2010, ISSN: 1352-2310. DOI: 10.1016/j.atmosenv.2010.01.002. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1352231010000099 (visited on 02/13/2026).

[4] E. Irankunda, M. Menendez, B. Khan, F. Paparella, and O. Pauluis, "Systematic review and meta-analysis of urban air quality in the arabian peninsula," *Atmosphere*, vol. 16, no. 8, p. 990, Aug. 20, 2025, ISSN: 2073-4433. DOI: 10.3390/atmos16080990. [Online]. Available: https://www.mdpi.com/2073-4433/16/8/990 (visited on 02/13/2026).

[5] IPCC, "Climate change 2022: Mitigation of climate change. contribution of working group III to the sixth assessment report of the intergovernmental panel on climate change," Intergovernmental Panel on Climate Change, Geneva, Switzerland, 2022. DOI: 10.1017/9781009157926. [Online]. Available: https://www.ipcc.ch/report/ar6/wg3/.

[6] F. Creutzig, P. Jochem, O. Y. Edelenbosch, *et al.*, "Transport: A roadblock to climate change mitigation?" *Science*, vol. 350, no. 6263, pp. 911–912, Nov. 20, 2015, ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aac8033. [Online]. Available: https://www.science.org/doi/10.1126/science.aac8033 (visited on 02/13/2026).

[7] N. D. of City Planning. "Population," Population - Department of City Planning. (Jul. 2024), [Online]. Available: https://www.nyc.gov/content/planning/pages/planning/population (visited on 02/13/2026).

[8] O. US EPA. "Greenhouse gas emissions from a typical passenger vehicle." (Jan. 12, 2016), [Online]. Available: https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle (visited on 02/13/2026).

[9] N. D. of Health {and} Mental Hygiene. "Traffic and PM2.5 air pollution," Environment & Health Data Portal. (), [Online]. Available: https://a816-dohbesp.nyc.gov/IndicatorPublic/data-stories/traffic-and-air-pollution/ (visited on 02/13/2026).

[10] M. C. Pinto de Moura and D. Reichmuth, "Inequitable exposure to air pollution from vehicles in the northeast and mid-atlantic," Union of Concerned Scientists, Cambridge, MA, 1, 2019. [Online]. Available: https://www.ucsusa.org/resources/inequitable-exposure-air-pollution-vehicles.

[11] Y. Wang, H. Wang, B. Zhang, *et al.*, "High-resolution mapping of on-road vehicle emissions with real-time traffic datasets based on big data," *Atmospheric Chemistry and Physics*, vol. 25, no. 11, pp. 5537–5555, Jun. 4, 2025, ISSN: 1680-7316. DOI: 10.5194/acp-25-5537-2025. [Online]. Available: https://acp.copernicus.org/articles/25/5537/2025/ (visited on 12/29/2025).

[12] H. Carvalho, "The air we breathe: Differentials in global air quality monitoring," *The Lancet. Respiratory Medicine*, vol. 4, no. 8, pp. 603–605, Aug. 2016, ISSN: 2213-2619. DOI: 10.1016/S2213-2600(16)30180-1.

[13] T.-Y. Lin, M. Maire, S. Belongie, *et al.*, "Microsoft COCO: Common objects in context," arXiv, Feb. 21, 2015. DOI: 10.48550/arXiv.1405.0312. arXiv: 1405.0312. [Online]. Available: http://arxiv.org/abs/1405.0312 (visited on 03/01/2026).